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International AS and A-level

Biology (9610)
BL04 Control
Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL BIOLOGY (9610) BL04 JUNE 2024

The paper produced a range of marks from 7 to 68 (out of 75) and the mean mark of 42.9 which was up by 7.1 marks from BL04 in the summer of 2023. There were many excellent answers across the paper and there seemed to be a general improvement in that students expressed their ideas in a clearer and more concise way. Despite this, students should still be reminded to read the command words in questions carefully and follow what is required. Topics in which students demonstrated good knowledge included: stem cells and how they can be used to treat human disorders (**01.1**, **01.2** and **01.3**), myelinated neurones and saltatory conduction (**04.2**), the second messenger model and adrenaline (**05.3** and **05.4**). Knowledge was less secure on the use of restriction enzymes (**02.1**) and how bacteria can produce human proteins (**06.1**). Many students also found it hard applying knowledge of actin and myosin and the mechanism of contraction to the context in **03.3**. Overall, the performance on mathematically based questions was mixed, with better performance in **04.3** than **03.2**

QUESTION 01

01.1 was answered well, with just under 70% of students gaining the two marks. Where students failed to gain a mark, it was usually for an incorrect definition of pluripotent eg cells that can develop into only some types of cells.

Question **01.2** gave a wide range of possible answers and just under 85% of students gained at least one mark. Diabetes was most commonly used but many other valid disorders were given such as leukaemia, Alzheimer's disease and Parkinson's disease.

Students answered **01.3** well and just under 60% gained full marks. All the marking points were seen but the ideas of no embryos being destroyed and low risk of rejection were by far the most common answers. Where marks were not awarded, it was usually for comments such as the use of iPSCs are more ethical without any further qualification.

Students found question **01.4** slightly more challenging, with around 50% gaining one mark and only around 25% gaining full marks. The ideas of drug testing or studying organ function were most commonly seen. Many answers stated that these mini organs could be used for organ transplant but this was idea ignored.

QUESTION 02

In question **02.1** students had to explain how different restriction enzymes would produce different length flanking regions. Although just over 50% gained one mark, very few achieved full marks. Many answers referred to the idea that different restriction enzymes would cut DNA at different recognition sequences but then failed to explain that these sites would be found at different distances from the gene.

Around a third of students gained full marks in **02.2** with some good descriptions of the graph in Figure 1. Others just gave vague descriptions eg ‘the greater the flanking region the greater the recombination efficiency’ with no further detail added.

Over 50% of students achieved the mark in question **02.3**. Other answers often had vague statements eg not referring to where the overlap / non-overlap of error bars occurred. There were a number of responses that stated that the results are significant (or not significant) rather than emphasising the differences between the results at different flanking region sizes.

QUESTION 03

In question **03.1** just over 75% gained one or both marks for correctly labelling the locations of ATPase and tropomyosin. Where marks were not awarded, it was typically for mixing up the locations or for not labelling the diagram accurately enough.

Around 45% correctly calculated the length of the muscle fibre and achieved full marks in **03.2**. There were numerous errors observed but one common mistake was in not using the scale bar to calculate the sarcomere length when relaxed. There were also errors in converting between units.

In **03.3** around 45% of student achieved one mark but less than 10% gained full marks. Many students just described how Figure 3 would look different, rather than explaining the difference when the myofibril is contracted. Some answers just had a general description of how a sarcomere appears different when contracted eg the H-zone narrows but this was not required. A common misconception in several answers was that the dots already shown in Figure 3 would just become thicker as the myofibril contracts. Many answers did not refer to actin or myosin (just thin / thick filaments).

Question **03.4** was a good discriminator and was generally answered well. The majority of answers gained at least one mark and around 45% achieved 5 or more marks. All the marking points were seen, although references to calcium ions activating ATPase were less common. Fewer students than expected referred to actinomyosin bridge formation but descriptions of this were allowed. Some students confused the role of ATPase in producing ATP, rather than hydrolysing ATP.

QUESTION 04

Many students achieved one mark in **04.1** for the idea that the threshold value had been met in test E (or conversely not met in test F). Just over 20% of students went a stage further and included a comparative statement about the entry of sodium ions. Some answers just gave a description of an action potential without relating this detail to the context of the question.

Question **04.2** was answered well, with most answers achieving one mark. Typically this was for references to saltatory conduction in myelinated neurones. Over 55% of students gave the additional detail that depolarisation can only take place at the nodes or that the myelin sheath prevents ion movement.

Around 65% of students gained two marks for the calculation in **04.3** Less than 20% of students achieved full marks by realising that the final step in the calculation involved dividing by 2 to get the myelin thickness.

Just over 70% of students gained two or more marks in question **04.4**. Some students misinterpreted the g-ratio and incorrectly stated that the Cyfip mutation would increase the myelin thickness (so not achieving the second marking point). Many answers correctly described the trend in Figure 8 relating to myelin thickness and conduction speed or stated that the mutation would lead to lower conduction speed.

QUESTION 05

05.1 was answered very well, with over 75% of students correctly describing negative feedback.

Students found **05.2** more challenging, with only 30% achieving full marks and just under 40% achieving one mark. Many answers described how corrective mechanisms would remain on but didn't mention that this would result in a further deviation from the set point. Some students just gave situations where positive feedback occurs but didn't then explain why this feedback is rare.

In question **05.3** students did refer to Figure 9 and this helped in the descriptions of how adrenaline increases blood glucose concentration. The question discriminated well and there were many excellent answers demonstrating good understanding of this content. Just over 60% of students gained 3 or more marks and close to 20% achieved full marks. Where marks were not gained, it was typically for not referring to adenyl cyclase or protein kinase. Some didn't refer to glycogen being hydrolysed to glucose or incorrectly referred to glycolysis instead of glycogenolysis.

Question **05.4** was answered well, with just under 50% of students achieving full marks and just under 40% getting one mark. Most of these answers referred to ATP being the product of respiration within the mitochondria. Some answers didn't make it clear that cAMP is made from ATP or other answers just gave vague references to ATP or to other factors not required for this answer eg energy for active transport.

Around 45% achieved both marking points in **05.5** and just over 30% gained one mark. Some students just repeated the question and stated that oestrogen is lipid-based without any further qualification.

QUESTION 06

Question **06.1** was not answered very well, with less than 10% achieving full marks and just over 35% gaining one mark. The majority of answers did not include the idea that the genetic code is universal and instead many referred to DNA as universal or that the bases are the same in all organisms which was insufficient for this first marking point. Many answers also focused on the human gene being copied when the bacteria divide, but with no detail as to how the bacteria are then able to produce the human protein.

Question **06.2** proved to be a good discriminator, with around 65% achieving at least two marks and 25% achieving full marks. It was evident that many students could name the two enzymes but the descriptions of stages 2 and 3 were often unclear or lacking detail. It was noticed that the term complementary DNA (cDNA) was not seen very often.

06.3 was answered well with over 50% gaining one mark, typically for referring to the idea of complementary sticky ends being produced. Around 25% achieved full marks for explaining that the same restriction enzyme would cut the DNA at specific recognition sites.

Just under 60% of students achieved full marks in **06.4**. Unfortunately, some students just stated the total number of colonies eg 5 colonies with no plasmids with antibiotic resistance genes. Some stated 1, 3, 11 and 12 for the colonies with the human gene not realising that the human gene would have disrupted the tetracycline gene.

In **06.5** just over 40% of students gained the one mark and this was typically for expressing the idea that the process is more time consuming or that replica plating is required. A number of answers referred to the bacteria mutating, which was not relevant to this question.

Question **06.6** was answered very well, with just under 85% of students gaining the one mark. Most of these students referred to a fluorescent gene marker or the GFP gene.

QUESTION 07

07.1 was answered well with around 75% of students achieving one or more marks. In some responses, key details were missing eg the arrival of an action potential in causing the influx of calcium ions. A number of answers incorrectly referred to the release of vesicles into the synaptic cleft as opposed to the release of the neurotransmitter.

Over 50% gained one mark in **07.2** and just under 25% achieved full marks. Most of these answers had the correct idea for the second marking point and ‘prevents continued stimulation’ was seen frequently. It was idea that serotonin would not continue binding to receptors that was often missed.

Around 90% gained at least one mark in **07.3** and this was typically for recognising that group Q was a control group to allow the other groups to be compared against. Some just stated ‘to act as a control’ without further qualification. Only around 30% went a stage further and explained that it would show that any changes in the other groups would therefore be due to MDMA.

Around 40% of students gained one mark in **07.4** and 30% achieved full marks. Figure 14 shows that groups R, S and T have a significant difference compared to group Q and so the first marking point needed the idea that MDMA caused a significant increase in serotonin concentration. Many answers commented that there was no significant difference between groups R – T.

There were some good answers in **07.5** with around 30% achieving full marks. Where marks were not awarded, it was usually for vague statements such as ‘MDMA acts like a competitive inhibitor’ but with no further qualification. Some students confused the role of the SERT despite the information given that they reabsorb serotonin from the cleft.

07.6 discriminated well and there were some excellent answers to this evaluation question. All the marking points were observed but it was more common to see mark points 1, 2, 5 and 7. A good number of students stated that no statistical test has been carried out despite being given a correlation coefficient and P values. Some students also confused the negative correlation coefficient as meaning that the correlation isn’t significant or that correlation coefficient is larger than the P value, so the correlation isn’t significant. Many students just stated a ‘small sample size’ or ‘only male rats’ without any further qualification.